

CS1010 Discrete Mathematics for Computer Science

Counting

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Why Do We Need Counting?

- ▶ A computer often has to deal with a huge collection of **possible states, inputs, configurations, or outcomes**.
- ▶ Before designing an algorithm, we may want to know:
 - ▷ How large is the space of possible inputs?
 - ▷ How many configurations can a system have?
 - ▷ How many candidate solutions are there?
 - ▷ Can we afford to examine all possibilities?
- ▶ Thus, a fundamental question in CS: **How many possibilities are there?**

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Counting is the mathematical way of determining the size of a finite possibility space.

It is therefore fundamental to (along with many other applications):

- ▶ analyze the **complexity** of algorithms,
- ▶ understand **search spaces** and exhaustive search,
- ▶ estimate the size of **state spaces**,
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The Basics of Counting

The Product Rule

- ▶ Consider a task that can be broken down into two tasks.
 - ▷ The first task can be done in n_1 ways.
 - ▷ The second task can be done in n_2 ways.
- ▶ Then there are $n_1 \cdot n_2$ ways to do the task.
- ▶ Think from a cartesian product perspective: There are two sets
 - ▷ The first set contains all the ways to do the first task.
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Product Rule in Terms of Sets

Q. Let $A_1, A_2, \dots, A_m$ be finite sets, then what is the number of elements in the Cartesian product of these sets?

A. Choosing an element in the Cartesian product $A_1 \times A_2 \times \dots \times A_m$ is done by choosing an element in A_1 , an element in A_2 , and so on.

$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_m|$$

Q. How many bit strings of length 12 are there?

A. Since each of the 12 bits is either 0 or 1, the answer is $2^{12} = 4096$.

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Q. Suppose IITH student ID consists of two uppercase English letters followed by four digits. How many different student IDs are possible?

A. Each letter has 26 choices and each digit has 10 choices. Then,

$$\underbrace{26 \times 26}_{\substack{\text{26 choices} \\ \text{for each letter}}} \times \underbrace{10 \times 10 \times 10 \times 10}_{\substack{\text{10 choices} \\ \text{for each digit}}} = 6,760,000$$

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A. $26 \times 26 \times 26 \times 10 \times 10 = 1,757,600$



Example: Social Security Number

Q. A simplified Social Security Number is represented by



- ▶ *S*: Sex (3 choices)
- ▶ *YY*: Year of birth (100 choices)
- ▶ *MM*: Month of birth (12 choices)
- ▶ *DDD*: Department code (100 choices)
- ▶ *BBB*: Birth order (1000 choices)

How many different simplified French Social Security Numbers are possible?

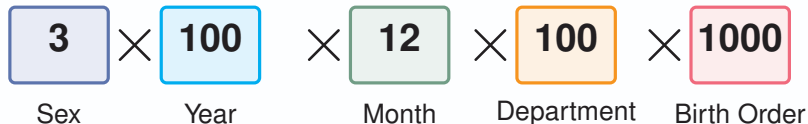
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Counting Functions

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A. Each of the m element of the domain can be mapped to any of the n elements of the codomain. Thus, the total number of functions: $n \cdot n \cdot \dots \cdot n = n^m$.

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Q. Number of injective functions from a set with m elements to one with n elements?

A. For the 1st element, there are n choices. For the 2nd element, there are $n - 1$ choices, and so on. Thus, in total: $n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - m + 1)$.

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Q. What is the number of different subsets of a finite set S ?

A. Think for each of the elements of S . Each element can be in the subset or not in the subset: two choices for each element. Thus, the total number of subsets is $2^{|S|}$.

The Sum Rule

- ▶ Consider a task can be done either in one of n_1 ways or in one of n_2 ways
 - ▶ None of the set of n_1 ways is the same as any of the n_2 ways
 - ▶ Then there are $n_1 + n_2$ ways to do the task.
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Q. A university offers two types of scholarships: 25 Merit Scholarships and 18 Sports Scholarships. No student is eligible for both scholarships. How many scholarships are available?

A. Choose either 25 Merit Scholarships or 18 Sports Scholarships. Hence, $25+18=43$ scholarships are available.

Q. What is the size of $|A \cup B|$ when A and B are disjoint sets, i.e., $A \cap B = \emptyset$?

A. The size of $A \cup B$ is the sum of the sizes of A and B , i.e., $|A \cup B| = |A| + |B|$.



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Combining the Sum and Product Rules

- Q. A university generates a **course access code** for every student. The code
- ▶ is either **5**, **6**, or **7** characters long,
 - ▶ each character is either an uppercase English letter or a digit,
 - ▶ and the code must contain **at least one digit**.

How many different course access codes are possible?

- Q. Total $N = N_5 + N_6 + N_7$ where N_i denotes the number of valid codes of length i . For a fixed length i ,
- ▶ there are 36^i possible codes (26 letters + 10 digits),
 - ▶ among them, 26^i contain only letters.

Thus, $N = (36^5 - 26^5) + (36^6 - 26^6) + (36^7 - 26^7)$



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Subtraction Rule

- ▶ Consider a task can be done either in one of n_1 ways or in one of n_2 ways
- ▶ Then the total number of ways to do the task is $n_1 + n_2$ minus the number of common ways
- ▶ Note the Difference from **Sum Rule**

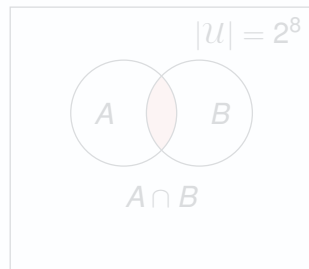
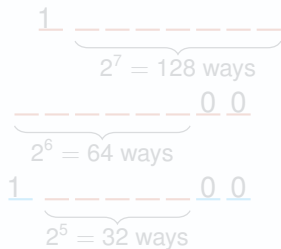
Principle of Inclusion-Exclusion

Q. What is the size of $A \cup B$?

A. $|A \cup B| = |A| + |B| - |A \cap B|.$

Example: Counting Bit Strings

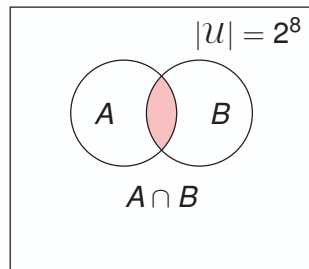
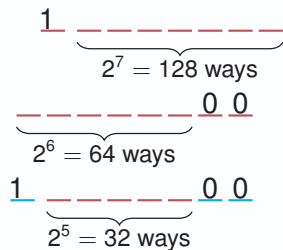
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A. $|A| = 2^7 = 128$, $|B| = 2^6 = 64$, and $|A \cap B| = 2^5 = 32$ so
 $|A \cup B| = 128 + 64 - 32 = 160$.

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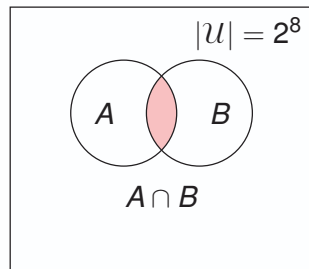
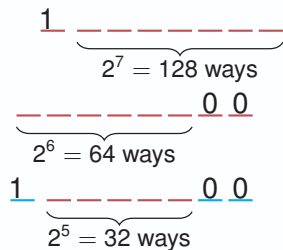
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Division Rule

In terms of counting: Sometimes we count the same object several times. If every object is counted exactly d times, divide the total count by d .

In terms of sets: If the finite set A is the union of n pairwise disjoint subsets each with d elements, then $n = |A|/d$.

In terms of functions: If f is a function from A to B , where both are finite sets, and for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$, then $|B| = |A|/d$.

Division Rule: An Example

Counting Unordered Pairs

Suppose we want to choose two students from $\{A, B, C, D\}$, where the order of selection does not matter.

- ▶ First count the ordered choices: $4 \times 3 \times 2 \times 1 = 12$.
- ▶ But each pair is counted twice:

(A, B) and (B, A)

represent the same pair.

- ▶ In fact, every unordered pair corresponds to exactly $d = 2$ ordered pairs.
- ▶ By the division rule, the number of unordered pairs is

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- A.* Therefore, by the division rule, there are $24/4 = 6$ different seating arrangements.



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A. $(333 - 33) - (83 - 8) = 300 - 75 = \boxed{225}$

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Q. How many positive integers between 100 and 999 inclusive are divisible by 3 but not by 4?

A. $(333 - 33) - (83 - 8) = 300 - 75 = \boxed{225}$

Q. How many positive integers between 100 and 999 inclusive are divisible by either 3 or 4?



Counting: Questions and Answers

Q. How many positive integers ≤ 100 are divisible either by 4 or by 6?

A.

$$\left\lfloor \frac{100}{4} \right\rfloor + \left\lfloor \frac{100}{6} \right\rfloor - \left\lfloor \frac{100}{12} \right\rfloor = 25 + 16 - 8 = \boxed{33}$$

Q. How many positive integers between 100 and 999 inclusive are divisible by 7?

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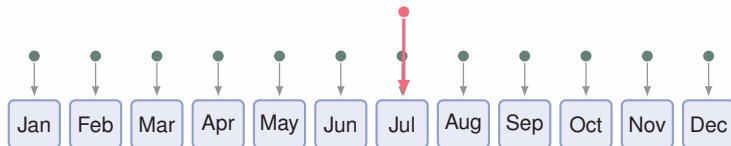
Q. How many positive integers between 100 and 999 inclusive are divisible by either 3 or 4?

A. $300 + 225 - 75 = \boxed{450}$

The Pigeonhole Principle

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If a class has **13 students**, then at least two students born in the same month



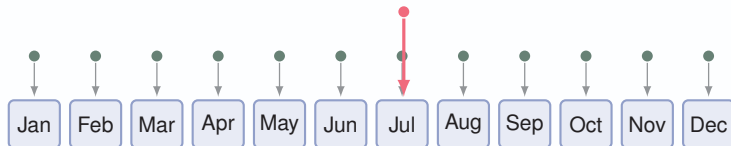
Pigeonhole Principle

If $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof (by contradiction): Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k . This contradicts the statement that we have $k + 1$ objects.

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Examples

Q. Can you define a one-to-one function from a set with $k + 1$ elements to a set with k elements?

A. The answer is **no**.

Q. Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.

A. Consider the $n + 1$ integers $1, 11, 111, \dots, \underbrace{11 \dots 1}_{n+1 \text{ ones}}$.

- ▶ When divided by n , there are only n possible remainders $\{0, 1, \dots, n - 1\}$.
- ▶ By the **Pigeonhole Principle**, two of these numbers have the same remainder.

Suppose they are $\underbrace{111 \dots 1}_a$ and $\underbrace{111 \dots 1}_b$, and $a > b$.

- ▶ Then $\underbrace{111 \dots 1100 \dots 0}_{\text{only 1s and 0s}} = \underbrace{111 \dots 1}_a - \underbrace{111 \dots 1}_b$ is divisible by n .

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Application of the Pigeonhole Principle

Q. Prove that if $n^2 + 1$ points are placed inside (or on the boundary of) an equilateral triangle of side length 1, then at least two of the points are at distance at most $\frac{1}{n}$.

A. Proof: Partition the triangle into n^2 equilateral triangles, each of side length $\frac{1}{n}$.

- ▶ These small triangles are the **boxes**.
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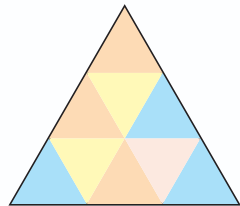
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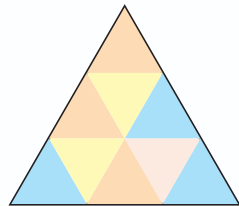
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The Generalized Pigeonhole Principle

Generalized Pigeonhole Principle

If N objects are placed into k boxes, then at least one box contains $\left\lceil \frac{N}{k} \right\rceil$ or more objects.

Proof (by contradiction)

- ▶ Assume every box contains at most $\left\lceil \frac{N}{k} \right\rceil - 1$ objects.
- ▶ Then the total number of objects is at most $k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right)$.
- ▶ Since $\left\lceil \frac{N}{k} \right\rceil < \frac{N}{k} + 1$, we obtain

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\frac{N}{k} + 1 - 1 \right) = N.$$

- ▶ This contradicts the assumption that there are exactly N objects.



Pigeonhole Principle: Questions and Answers

- Q. Show that if f is a function from S to T , where S and T are finite sets with $|S| > |T|$, then f is not one-to-one.

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- Q. How many ordered pairs of integers (a, b) are needed to guarantee that there are two ordered pairs (a_1, b_1) and (a_2, b_2) such that

$$a_1 \bmod 5 = a_2 \bmod 5 \quad \text{and} \quad b_1 \bmod 5 = b_2 \bmod 5?$$

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- A.* There are 5 possible residues for each coordinate, giving

$$5 \cdot 5 = 25$$

possible residue pairs. Therefore,

26.

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At most one number can be selected from each pair without obtaining a sum of 7. Thus, 4.

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A. Each point has one of $2^3 = 8$ possible parity patterns: $(x_i \bmod 2, y_i \bmod 2, z_i \bmod 2)$. Since there are 9 points and only 8 patterns, two points have the same parity pattern. Hence their midpoint has integer coordinates.



Increasing and Decreasing Subsequences

2. Show that if there are 101 people of different heights standing in a line, it is possible to find 11 people in the order they are standing in the line with heights that are either increasing or decreasing.

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- A. But there are 101 people. So, two people must have the same pair (I_i, D_i) .
- A. This is impossible: for two people $i < j$, either $h_i < h_j \Rightarrow I_j > I_i$, or $h_i > h_j \Rightarrow D_j > D_i$.



Thank You for your
attention!

Any questions?